

GENIE

Google Gemini Powered Learning Assistant

Project Design Phase

Designing the architecture, workflow, and user interaction

Project Phase Documentation

This document covers the activities, outputs, and considerations for this phase of the Genie project.

1. System Architecture

Genie – Project Design Phase

- Genie can be designed as a simple client-application and AI-service workflow.
- Main components: User Interface, Application Logic, Gemini API/Model, and Response Display.
- The user submits a prompt through the interface.
- The application validates and sends the prompt to the Gemini service.
- The generated response is returned to the application and displayed to the user.

Aspect	Details
Purpose	Designing the architecture, workflow, and user interaction
Project	Genie: Google Gemini Powered Learning Assistant
Technology	Python + Google Gemini integration
Target Users	Students / learners

2. User Interface & Interaction Design

Genie – Project Design Phase

- Provide a clear input area for the student's question.
- Provide a visible action such as Ask, Generate, or Submit.
- Display the AI response in a readable section.
- Use simple labels and navigation so that a student can understand the interface quickly.
- Keep the layout uncluttered and focused on the learning interaction.

Key Note: The exact implementation can be adjusted based on the final project code and available Gemini API configuration.

3. Data Flow & Design Considerations

Genie – Project Design Phase

- Input: student's learning question or prompt.
- Processing: application prepares the prompt and communicates with Gemini.
- AI generation: Gemini interprets the prompt and generates a response.
- Output: the application displays the response to the student.
- Design considerations include error handling, API-key security, response readability, and responsible use of generated information.

Phase Output: A clear, reviewable result that can be carried forward to the next project phase.